# Utilize total angular momentum of twisted light to control gray exciton transition rate

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# Laguerre-Gaussian (LG) Beam

LG Beam is a set of solutions for paraxial Helmholtz equation with various ansatzes, which carrying topological charge $\ell$.

**Used ansatzes**:

| Monochronic ansatz: | Transverse ansatz: |
|---|---|
| $\boldsymbol{A}(\boldsymbol{r},t) = \mathrm{Re}\left[\boldsymbol{A}(\boldsymbol{r})e^{-i\omega t}\right]$ | $\boldsymbol{A}(\boldsymbol{r}) = \hat{\boldsymbol{\epsilon}} A(\boldsymbol{r})$ |
| $q_0 = \omega/c$ is wavenumber of light | $\hat{\boldsymbol{\epsilon}}$ is polarization with unit modules |

Paraxial wave ansatz:

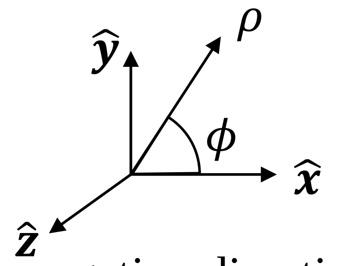
$$A(\boldsymbol{r}) = u(\boldsymbol{r})e^{iq_0 z}$$

where

$\nabla_{\parallel}^2 u(\boldsymbol{r}) \gg \frac{\partial^2}{\partial^2 z}u(\boldsymbol{r})$, and

$\frac{\partial}{\partial z}u(\boldsymbol{r}) \gg \frac{\partial^2}{\partial^2 z}u(\boldsymbol{r})$.

$\boldsymbol{r} = (\boldsymbol{r}_{\parallel}, z) = (\rho, \phi, z)$




(propagation direction)

**Paraxial Helmholtz equation** (from Maxwell's equation)

$$\nabla_{\parallel}^2 u(\boldsymbol{r}) + 2iq_0 u(\boldsymbol{r}) = 0$$

+long Rayleigh range approximation

**Solutions of LG Beam**

$$\boldsymbol{A}^{\hat{\epsilon},\ell,p;L}(\boldsymbol{r},t) = \hat{\boldsymbol{\epsilon}} u_{\ell,p}(\boldsymbol{r}_{\parallel})e^{iq_0 z}e^{-i\omega t} = \hat{\boldsymbol{\epsilon}} A_0 \sqrt{\frac{2p!}{\pi(|\ell|+p!)}} L_p^{|\ell|}\left(\frac{2\rho^2}{W_0^2}\right)\left(\frac{\sqrt{2}\rho}{W_0}\right)^{|\ell|} e^{-\frac{\rho^2}{W_0^2}} e^{i\ell\phi}e^{iq_0 z}e^{-i\omega t}$$

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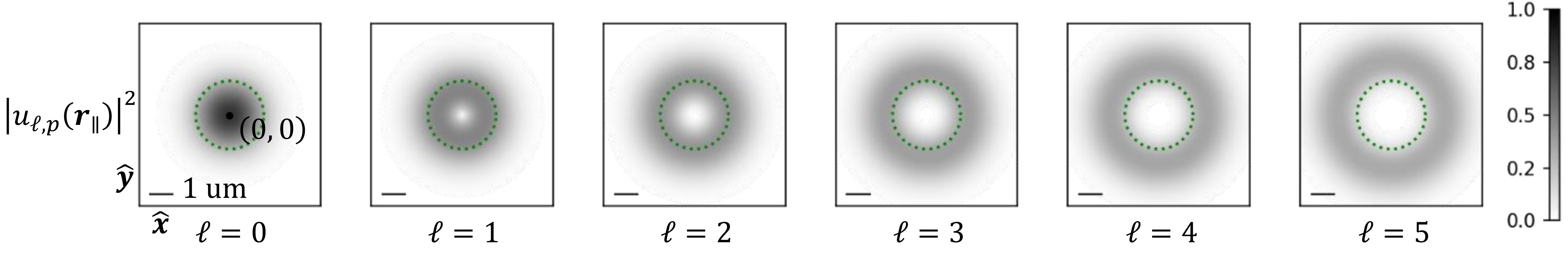


$W_0 = 1.5$ um is beam waist, $L_p^{|\ell|}$ is the generalized Laguerre polynomial

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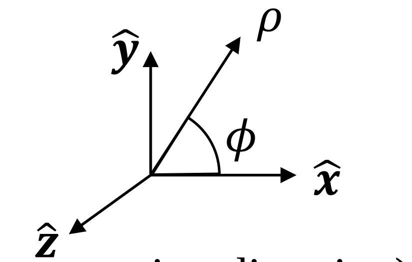

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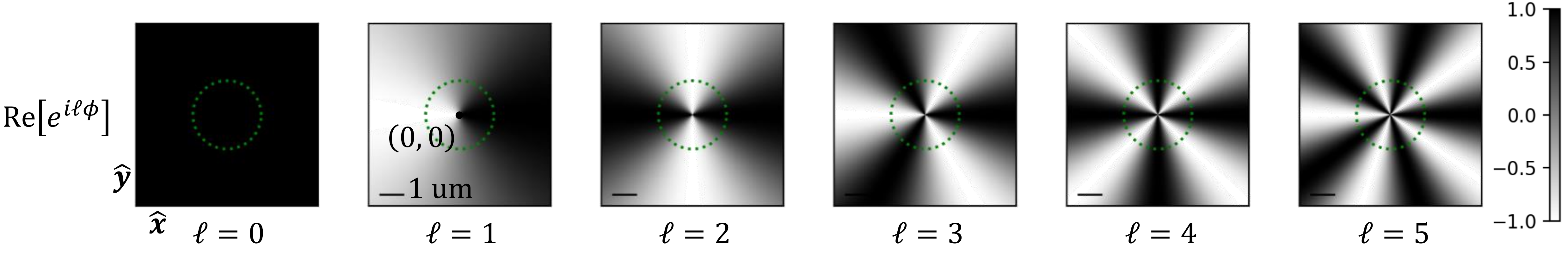


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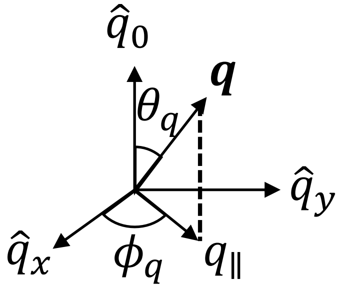
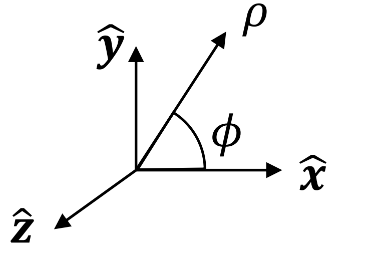
# Angular Spectrum Representation under Coulomb gauge

Obtain LG beams in the angular spectrum representation through Fourier transforms, and further turn it into Coulomb gauge to meet the requirement in light-matter inter action theory.

$$\boldsymbol{r} = (\boldsymbol{r}_\parallel, z) = (\rho, \phi, z)$$




$$\boldsymbol{q} = (\boldsymbol{q}_\parallel, q_0) = \left(q_x, q_y, q_0\right)$$

**Defined Fourier transform, and Inverse Fourier transform** :

$$u_{\ell p}(\boldsymbol{r}_\parallel) = \frac{1}{(2\pi)^2}\int d^2\boldsymbol{q}_\parallel F_{\ell p}(\boldsymbol{q}_\parallel)\, e^{i\boldsymbol{q}_\parallel\cdot\boldsymbol{r}_\parallel}\text{, and } F_{\ell p}(\boldsymbol{q}_\parallel) = \int d^2\boldsymbol{r}_\parallel u_{\ell p}(\boldsymbol{r}_\parallel)\, e^{-i\boldsymbol{q}_\parallel\cdot\boldsymbol{r}_\parallel}$$

**Relation between gauges:**
(from Maxwell's equation)

$$\boldsymbol{A}^{\sigma,\ell,p;C}(\boldsymbol{r}) = \boldsymbol{A}^{\sigma,\ell,p;L}(\boldsymbol{r}) + \frac{\nabla\left(\nabla\cdot\boldsymbol{A}^{\sigma,\ell,p;L}(\boldsymbol{r})\right)}{q_0^2}$$

**LG beam in angular spectrum representation under Coulomb gauge:**

$$\boldsymbol{\mathcal{A}}^{\sigma,\ell,p;C}(\boldsymbol{q}_\parallel) \approx \tilde{F}_{|\ell|,p}(q_\parallel)e^{i\ell\phi_q}\hat{\boldsymbol{\epsilon}} - \frac{\sin\theta_{\boldsymbol{q}}}{\sqrt{2}}\tilde{F}_{|\ell|,p}(q_\parallel)e^{i(\ell+\sigma)\phi_{\boldsymbol{q}}}\hat{\boldsymbol{z}}$$

$$\tilde{F}_{|\ell|,p=0}(q_\parallel) = (-i)^{|\ell|}\frac{A_0}{\Omega}\sqrt{\frac{2\pi}{|\ell|!}}W_0^2\left(\frac{q_\parallel W_0}{\sqrt{2}}\right)^{|\ell|} e^{-\frac{q_\parallel^2 W_0^2}{4}}$$

$\hat{\boldsymbol{\epsilon}} = \frac{1}{\sqrt{2}}(\hat{\boldsymbol{x}} \pm i\sigma\hat{\boldsymbol{y}})$, $\sigma = \pm 1$, and $\Omega$ is the area of the 2D material

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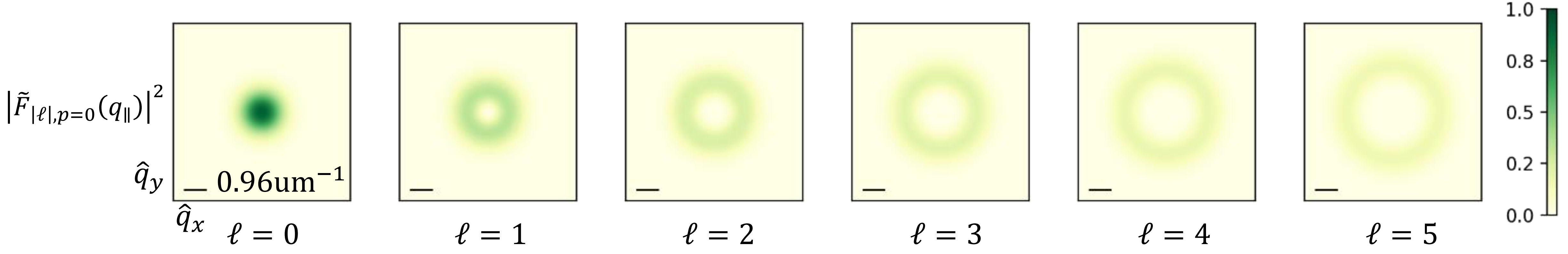

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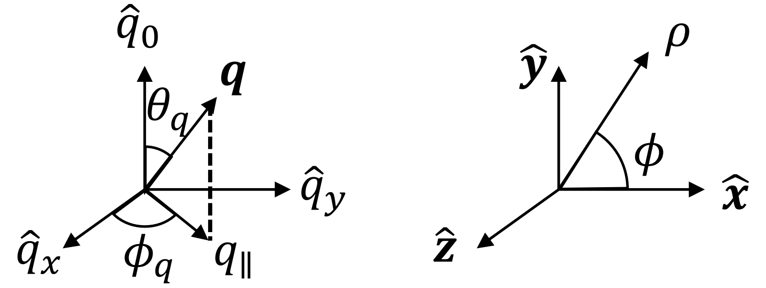


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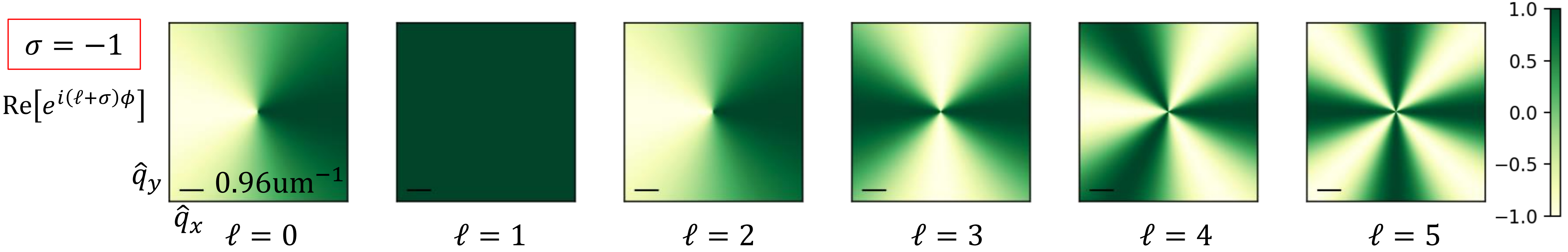


# Excitations of Excitons with Single Twisted Light

**Absorption transition rate** $\Gamma$ root from the Fermi's golden rule, describe the optical transition as from many body ground state to the exciton state with center-of-mass momentum $\boldsymbol{Q}$

$$\Gamma_{S,\boldsymbol{Q}}^{\sigma,\ell,p} = \frac{2\pi}{\hbar}\left|\widetilde{M}_{S,\boldsymbol{Q}}^{\sigma,\ell,p}\right|^2 \rho^L\left(\hbar\omega = E_{S,\boldsymbol{Q}}^X\right)$$

$$\widetilde{M}_{S,\boldsymbol{Q}}^{\sigma,\ell,p} \approx \frac{E_g}{2i\hbar}\boldsymbol{\mathcal{A}}^{\sigma,\ell,p;C}(\boldsymbol{Q}) \cdot \boldsymbol{D}_{S,\boldsymbol{Q}}^{X*}$$

$S = B+, B-, G$ for denoting upper bright, lower bright, and gray exciton band

$\boldsymbol{Q} = (Q, \phi_{\boldsymbol{Q}}) \approx \boldsymbol{q}_{\parallel}$ is the center-of-mass momentum of exciton

$\boldsymbol{D}_{S,\boldsymbol{Q}}^X$ is the transition dipole moment of an exciton state

**Optical matrix elements for** $p = 0$**:**

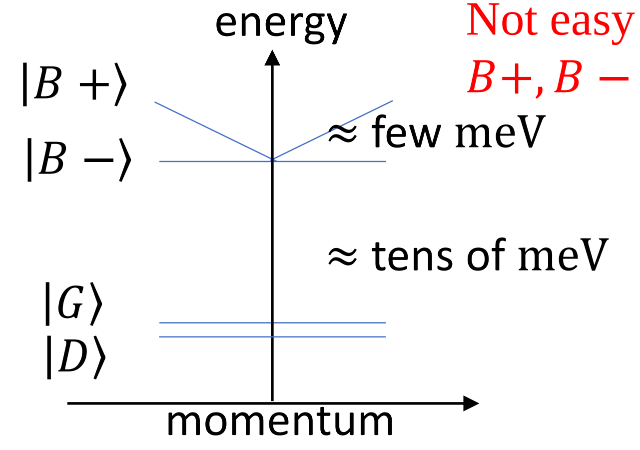
$$\widetilde{M}_{B+,\boldsymbol{Q}}^{\sigma,\ell,p=0} \approx \frac{E_g D_{B+}^X}{2\sqrt{2}i\hbar}\tilde{F}_{|\ell|,p=0}(Q)e^{i\boxed{(\ell+\sigma)}\phi_{\boldsymbol{Q}}}$$

$$\widetilde{M}_{B-,\boldsymbol{Q}}^{\sigma,\ell,p=0} \approx \sigma\frac{E_g D_{B-}^X}{2\sqrt{2}i\hbar}\tilde{F}_{|\ell|,p=0}(Q)e^{i\boxed{(\ell+\sigma)}\phi_{\boldsymbol{Q}}}$$

$$\widetilde{M}_{G,\boldsymbol{Q}}^{\sigma,\ell,p=0} \approx -\sin\theta_{\boldsymbol{Q}}\frac{E_g D_{G}^X}{2\sqrt{2}i\hbar}\tilde{F}_{|\ell|,p=0}(Q)e^{i\boxed{(\ell+\sigma)}\phi_{\boldsymbol{Q}}} \quad \sin\theta_{\boldsymbol{Q}} \equiv Q/\sqrt{Q^2+q_0^2}$$

Can't observe phase term

Illustration of W-based TMD exciton band structure




Not easy to observe $B+, B-$ saperately

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# Excitations of Excitons with Vector Vortex Beams

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$$\boldsymbol{\mathcal{A}}^{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel)$$
$$= \cos(\beta/2)\,\boldsymbol{\mathcal{A}}^{\sigma,\ell,p;C}(\boldsymbol{q}_\parallel) + e^{i\alpha}\sin(\beta/2)\,\boldsymbol{\mathcal{A}}^{\sigma',\ell',p;C}(\boldsymbol{q}_\parallel)$$
$$\sigma \neq \sigma', \ell \neq \ell'$$

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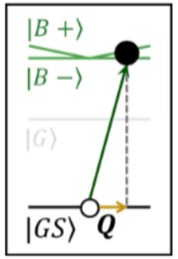
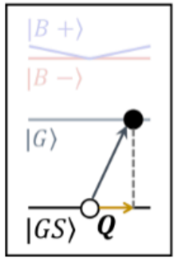



$$\sum_{S=B\pm}\left|\widetilde{M}_{S,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'}\right|^2 \approx$$

$$\left(\frac{E_g D_{B+}^{X}}{2\hbar}\right)^2\left[\left|\tilde{F}_{|\ell|}(Q)\right|^2\cos^2(\beta/2) + \left|\tilde{F}_{|\ell'|}(Q)\right|^2\sin^2(\beta/2)\right]$$




$$\left|\widetilde{M}_{G,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'}\right|^2 \approx$$

Total angular momentum appears in $\Delta J$

$$\left(\frac{E_g D_G^X \sin\theta_{\boldsymbol{Q}}}{2\sqrt{2}\hbar}\right)^2 \begin{cases} \left|\tilde{F}_{|\ell|}(Q)\right|^2\cos^2(\beta/2) + \left|\tilde{F}_{|\ell'|}(Q)\right|^2\sin^2(\beta/2) \\ +\mathrm{Im}\left[\tilde{F}_{|\ell|}(Q)\right]\mathrm{Im}\left[\tilde{F}_{|\ell'|}(Q)\right]\sin(\beta)\cos(\boxed{\Delta J}\phi_{\boldsymbol{Q}}+\alpha) \\ +\mathrm{Re}\left[\tilde{F}_{|\ell|}(Q)\right]\mathrm{Re}\left[\tilde{F}_{|\ell'|}(Q)\right]\sin(\beta)\cos(\boxed{\Delta J}\phi_{\boldsymbol{Q}}+\alpha) \\ -\mathrm{Re}\left[\tilde{F}_{|\ell|}(Q)\right]\mathrm{Im}\left[\tilde{F}_{|\ell'|}(Q)\right]\sin(\beta)\sin(\boxed{\Delta J}\phi_{\boldsymbol{Q}}+\alpha) \\ +\mathrm{Im}\left[\tilde{F}_{|\ell|}(Q)\right]\mathrm{Re}\left[\tilde{F}_{|\ell'|}(Q)\right]\sin(\beta)\sin(\boxed{\Delta J}\phi_{\boldsymbol{Q}}+\alpha) \end{cases}$$

$$\Delta J = (\sigma' + \ell') - (\sigma + \ell)$$

# Excitations of Excitons with Vector Vortex Beams

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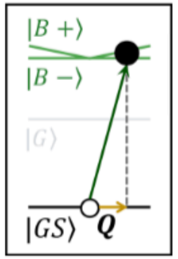
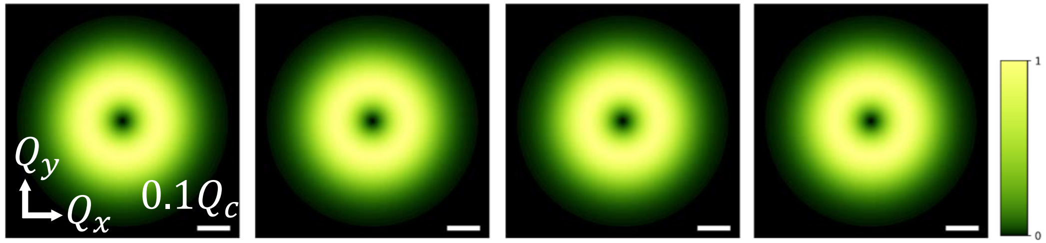
$$\sigma \neq \sigma', \ell \neq \ell'$$

**Optical matrix elements for** $p = 0$ **:**

$$\sum_{S=B\pm,\boldsymbol{Q}}\left|\widetilde{M}_{S,\boldsymbol{Q}}^{1,-1,-1,1}\right|^2 \approx$$

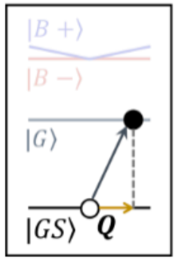


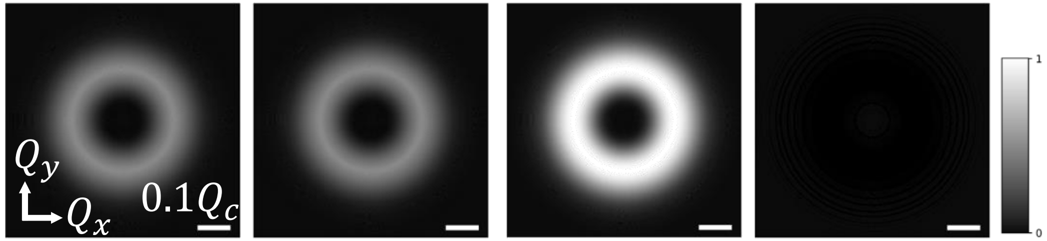




β = 0 | β = π | α = 0, β = π/2 | α = π, β = π/2

$$\left|\widetilde{M}_{G,\boldsymbol{Q}}^{1,-1,-1,1}\right|^2 \approx$$

Control with relative phase $\alpha$

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$$\boxed{\Delta J = (\sigma' + \ell') - (\sigma + \ell) = 0}$$

# Excitations of Excitons with Vector Vortex Beams

**Transition rate** (total)

$$\Gamma_S^{\sigma,\ell,\sigma',\ell'} = \sum_{\boldsymbol{Q}} \Gamma_{S,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'} \propto \sum_{\boldsymbol{Q}} \left| \widetilde{M}_{S,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'} \right|^2$$

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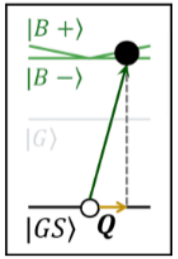
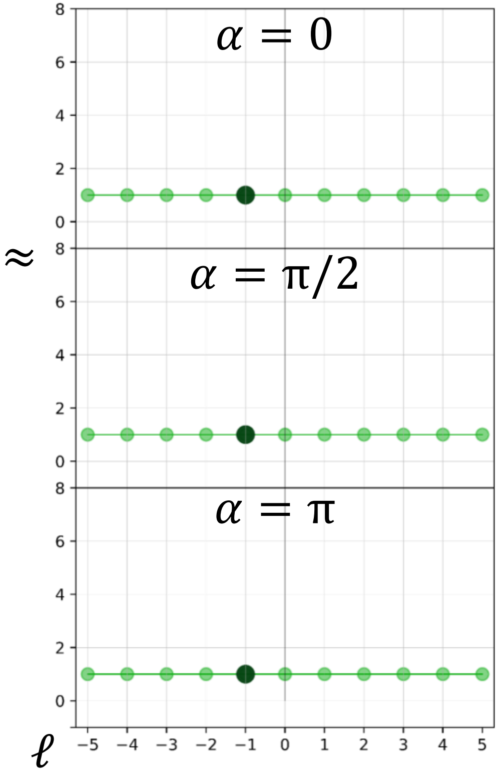
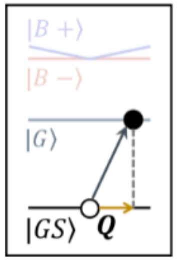



$$\Gamma_{B\pm}^{1,\ell,-1,-\ell}(\alpha, \beta = \pi/2) \propto \sum_{S=B\pm,\boldsymbol{Q}} \left| \widetilde{M}_{S,\boldsymbol{Q}}^{1,\ell,-1,-\ell} \right|^2 \approx$$

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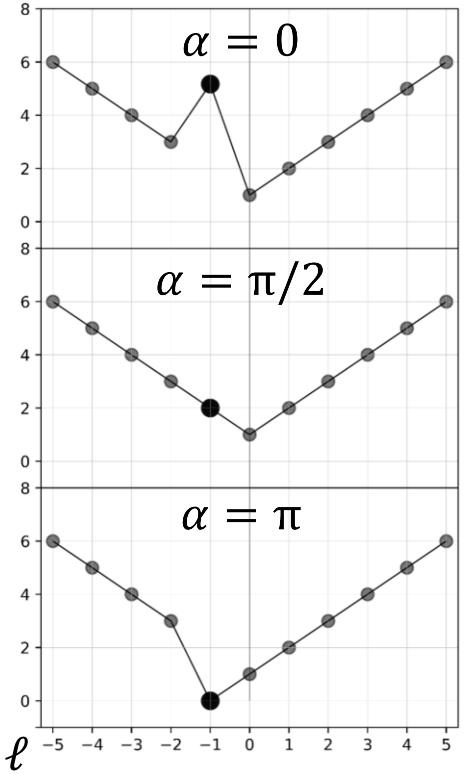


Only when $\ell = -1$
can we control with $\alpha$

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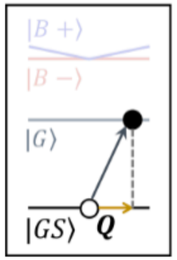
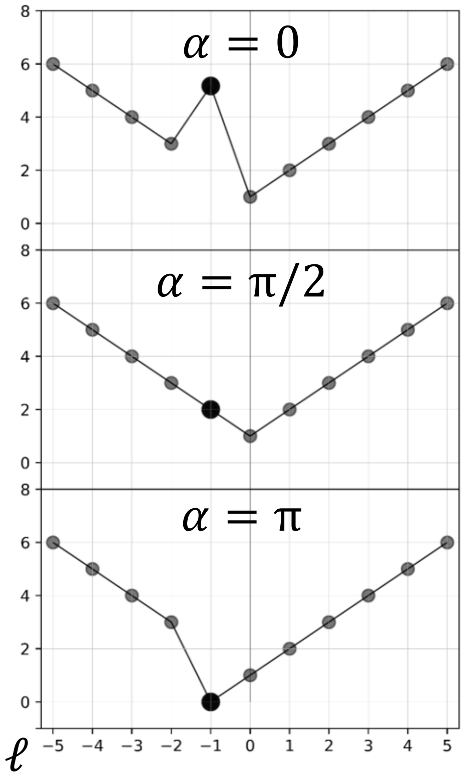



$$\Gamma_G^{1,\ell,-1,-\ell}(\alpha, \beta = \pi/2) \propto \left(1 + \frac{|\ell| + |\ell'|}{2}\right)\left[1 + \delta_{\Delta J,0}\left(\delta_{J,0} - \sqrt{\frac{|J|}{|J|+1}} \cos\alpha\right)\right]$$

$$J = +\sigma + \ell$$

$$J' = \sigma' + \ell'$$

Control $\Gamma_G^{1,\ell,-1,-\ell}$ with $\alpha$ when $J = J' = 0$ $(\ell = -1)$



# Thank you for listening

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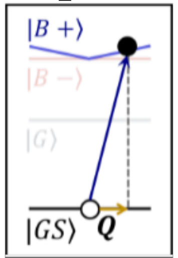
$$\boldsymbol{\mathcal{A}}^{\sigma,\ell,p;C}(\boldsymbol{q}_\parallel) \approx \tilde{F}_{|\ell|,p}(q_\parallel)e^{i\ell\phi_{\boldsymbol{q}}}\hat{\boldsymbol{\epsilon}} - \frac{\sin\theta_{\boldsymbol{q}}}{\sqrt{2}}\tilde{F}_{|\ell|,p}(q_\parallel)e^{i(\ell+\sigma)\phi_{\boldsymbol{q}}}\hat{\boldsymbol{z}}$$

$$\tilde{F}_{|\ell|,p=0}(q_\parallel) = \tilde{F}_{|\ell|}(Q) = (-i)^{|\ell|}\frac{A_0}{\Omega}\sqrt{\frac{2\pi}{|\ell|!}}W_0^2\left(\frac{q_\parallel W_0}{\sqrt{2}}\right)^{|\ell|} e^{-\frac{q_\parallel^2 W_0^2}{4}}$$

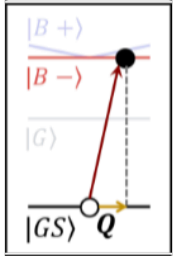
**Optical matrix elements for** $p = 0$




$$\tilde{M}_{B+,\boldsymbol{Q}}^{\sigma,\ell} \approx \frac{E_g D_{B+}^X}{2\sqrt{2}i\hbar}\tilde{F}_{|\ell|,p=0}(Q)e^{i(\ell+\sigma)\phi_{\boldsymbol{Q}}}$$

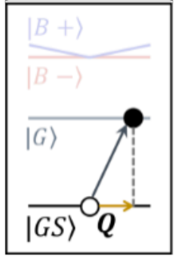
$$\boldsymbol{D}_{B+,\boldsymbol{Q}}^X = D_{B+}^X\left(\cos\phi_{\boldsymbol{Q}}\hat{\boldsymbol{x}} + \sin\phi_{\boldsymbol{Q}}\hat{\boldsymbol{y}}\right)$$

$$\boldsymbol{D}_{B-,\boldsymbol{Q}}^X = D_{B-}^X\left(-\sin\phi_{\boldsymbol{Q}}\hat{\boldsymbol{x}} + \cos\phi_{\boldsymbol{Q}}\hat{\boldsymbol{y}}\right)$$




$$\tilde{M}_{B-,\boldsymbol{Q}}^{\sigma,\ell} \approx \sigma\frac{E_g D_{B-}^X}{2\sqrt{2}i\hbar}\tilde{F}_{|\ell|,p=0}(Q)e^{i(\ell+\sigma)\phi_{\boldsymbol{Q}}}$$

$$\boldsymbol{D}_{G,\boldsymbol{Q}}^X = D_G^X\hat{\boldsymbol{z}}$$




$$\tilde{M}_{G,\boldsymbol{Q}}^{\sigma,\ell} \approx -\sin\theta_{\boldsymbol{Q}}\frac{E_g D_G^X}{2\sqrt{2}i\hbar}\tilde{F}_{|\ell|,p=0}(Q)e^{i(\ell+\sigma)\phi_{\boldsymbol{Q}}} \quad \sin\theta_{\boldsymbol{Q}} \equiv Q/\sqrt{Q^2+q_0^2}$$

# Excitations of Excitons with Vector Vortex Beams

**Transition rate** of absorption $\Gamma$ root from the Fermi's golden rule, describe the optical transition as from many body ground state to the exciton state with center-of-mass momentum $\boldsymbol{Q}$

$$\Gamma_{S,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'} = \frac{2\pi}{\hbar}\left|\widetilde{M}_{S,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'}\right|^2 \rho(\hbar\omega = E_{S,\boldsymbol{Q}}^{X})$$

$$M_{S,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'} \approx \cos(\beta/2)\, M_{S,\boldsymbol{Q}}^{\sigma,\ell} + e^{i\alpha}\sin(\beta/2)\, M_{S,\boldsymbol{Q}}^{\sigma',\ell'}$$

$$M_{S,\boldsymbol{Q}}^{\sigma,\ell} \approx \frac{E_g}{2i\hbar}\boldsymbol{\mathcal{A}}^{\sigma,\ell}(\boldsymbol{Q})\cdot \boldsymbol{D}_{S,\boldsymbol{Q}}^{X*}$$

**Vector Vortex Beams** is composed by two twisted light, and we denote it with each light's spin $\sigma$ and topological charge $\ell$

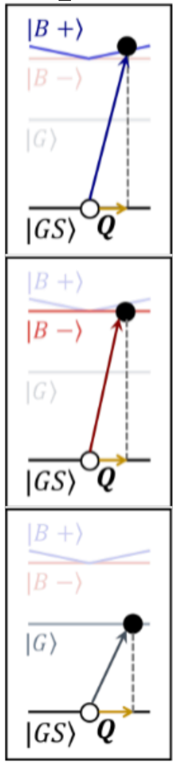
$$\boldsymbol{\mathcal{A}}^{\sigma,\ell,p;C}(\boldsymbol{q}_\parallel) \approx \tilde{F}_{|\ell|,p}(q_\parallel)e^{i\ell\phi_{\boldsymbol{q}}}\hat{\boldsymbol{\epsilon}} - \frac{\sin\theta_{\boldsymbol{q}}}{\sqrt{2}}\tilde{F}_{|\ell|,p}(q_\parallel)e^{i(\ell+\sigma)\phi_{\boldsymbol{q}}}\hat{\boldsymbol{z}}$$

$$\tilde{F}_{|\ell|,p=0}(q_\parallel) = \tilde{F}_{|\ell|}(Q) = (-i)^{|\ell|}\frac{A_0}{\Omega}\sqrt{\frac{2\pi}{|\ell|!}}W_0^2\left(\frac{q_\parallel W_0}{\sqrt{2}}\right)^{|\ell|} e^{-\frac{q_\parallel^2 W_0^2}{4}}$$

$$\boldsymbol{\mathcal{A}}^{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel) = \cos(\beta/2)\,\boldsymbol{\mathcal{A}}^{\sigma,\ell,p;C}(\boldsymbol{q}_\parallel) + e^{i\alpha}\sin(\beta/2)\,\boldsymbol{\mathcal{A}}^{\sigma',\ell',p;C}(\boldsymbol{q}_\parallel)$$

$$\Delta J = (\sigma' + \ell') - (\sigma + \ell)$$

**Optical matrix elements for $p = 0$ (from Oscar) :**

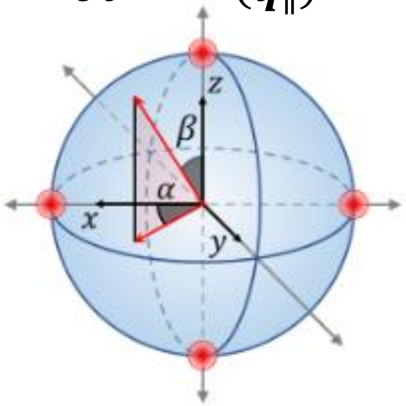



$$\left|\widetilde{M}_{B+,\boldsymbol{Q}}^{\sigma,\ell,\sigma'=-\sigma,\ell'}\right|^2 \approx \left(\frac{E_g D_{B+}^X}{2\sqrt{2}\hbar}\right)^2\Big[F_{|\ell|}^2(Q)\cos^2(\beta/2) + F_{|\ell'|}^2(Q)\sin^2(\beta/2) + F_{|\ell|}(Q)F_{|\ell'|}(Q)\sin(\beta)\cos\left(\Delta J\phi_{\boldsymbol{Q}} + \alpha - \frac{\pi}{2}(|\ell'| - |\ell|)\right)\Big]$$

$$\left|\widetilde{M}_{B-,\boldsymbol{Q}}^{\sigma,\ell,\sigma'=-\sigma,\ell'}\right|^2 \approx \left(\frac{E_g D_{B-}^X}{2\sqrt{2}\hbar}\right)^2\Big[F_{|\ell|}^2(Q)\cos^2(\beta/2) + F_{|\ell'|}^2(Q)\sin^2(\beta/2) - F_{|\ell|}(Q)F_{|\ell'|}(Q)\sin(\beta)\cos\left(\Delta J\phi_{\boldsymbol{Q}} + \alpha - \frac{\pi}{2}(|\ell'| - |\ell|)\right)\Big]$$

$$\left|\widetilde{M}_{G,\boldsymbol{Q}}^{\sigma,\ell,\sigma'=-\sigma,\ell'}\right|^2 \approx \left(\frac{E_g D_{G}^X \sin\theta_{\boldsymbol{Q}}}{2\sqrt{2}\hbar}\right)^2\Big[F_{|\ell|}^2(Q)\cos^2(\beta/2) + F_{|\ell'|}^2(Q)\sin^2(\beta/2) + F_{|\ell|}(Q)F_{|\ell'|}(Q)\sin(\beta)\cos\left(\Delta J\phi_{\boldsymbol{Q}} + \alpha - \frac{\pi}{2}(|\ell'| - |\ell|)\right)\Big]$$

$\boldsymbol{\mathcal{A}}^{\sigma,\ell,p;C}(\boldsymbol{q}_\parallel)$




$\boldsymbol{\mathcal{A}}^{\sigma',\ell',p;C}(\boldsymbol{q}_\parallel)$

# Excitations of Excitons with Vector Vortex Beams

**Transition rate** of absorption Γ root from the Fermi's golden rule, describe the optical transition as from many body ground state to the exciton state with center-of-mass momentum $\boldsymbol{Q}$

$$\Gamma_{S,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'} = \frac{2\pi}{\hbar}\left|\widetilde{M}_{S,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'}\right|^2 \rho\left(\hbar\omega = E_{S,\boldsymbol{Q}}^{X}\right)$$

$$M_{S,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'} \approx \cos(\beta/2)\, M_{S,\boldsymbol{Q}}^{\sigma,\ell} + e^{i\alpha}\sin(\beta/2)\, M_{S,\boldsymbol{Q}}^{\sigma',\ell'}$$

$$M_{S,\boldsymbol{Q}}^{\sigma,\ell} \approx \frac{E_g}{2i\hbar}\boldsymbol{\mathcal{A}}^{\sigma,\ell}(\boldsymbol{Q})\cdot \boldsymbol{D}_{S,\boldsymbol{Q}}^{X*}$$

**Vector Vortex Beams** is composed by two twisted light, and we denote it with each light's spin $\sigma$ and topological charge $\ell$

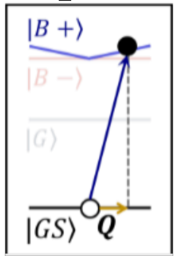
$$\boldsymbol{\mathcal{A}}^{\sigma,\ell,p;C}(\boldsymbol{q}_\parallel) \approx \tilde{F}_{|\ell|,p}(q_\parallel)e^{i\ell\phi_{\boldsymbol{q}}}\hat{\boldsymbol{\epsilon}} - \frac{\sin\theta_{\boldsymbol{q}}}{\sqrt{2}}\tilde{F}_{|\ell|,p}(q_\parallel)e^{i(\ell+\sigma)\phi_{\boldsymbol{q}}}\hat{\boldsymbol{z}}$$

$$\tilde{F}_{|\ell|,p=0}(q_\parallel) = \tilde{F}_{|\ell|}(Q) = (-i)^{|\ell|}\frac{A_0}{\Omega}\sqrt{\frac{2\pi}{|\ell|!}}W_0^2\left(\frac{q_\parallel W_0}{\sqrt{2}}\right)^{|\ell|}e^{-\frac{q_\parallel^2 W_0^2}{4}}$$

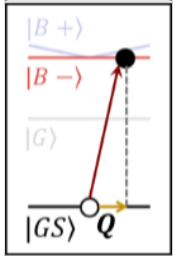
$$\boldsymbol{\mathcal{A}}^{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel) = \cos(\beta/2)\,\boldsymbol{\mathcal{A}}^{\sigma,\ell,p;C}(\boldsymbol{q}_\parallel) + e^{i\alpha}\sin(\beta/2)\,\boldsymbol{\mathcal{A}}^{\sigma',\ell',p;C}(\boldsymbol{q}_\parallel)$$

$$\Delta J = (\sigma' + \ell') - (\sigma + \ell)$$

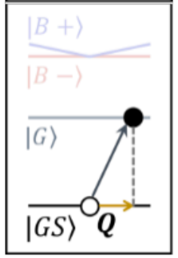
**Optical matrix elements for $p = 0$:**




$$\left|\widetilde{M}_{B+,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'}\right|^2 \approx \left(\frac{E_g D_{B+}^{X}}{2\sqrt{2}\hbar}\right)^2\left|\cos(\beta/2)\tilde{F}_{|\ell|}(Q) + \boxed{e^{-i(\Delta J\phi_{\boldsymbol{Q}}+\alpha)}}\sin(\beta/2)\tilde{F}_{|\ell'|}(Q)\right|^2$$

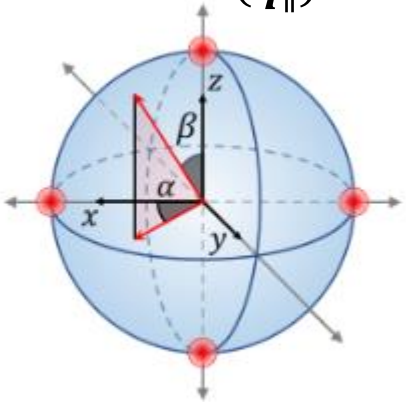



$$\left|\widetilde{M}_{B-,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'}\right|^2 \approx \left(\frac{E_g D_{B-}^{X}}{2\sqrt{2}\hbar}\right)^2\left|\cos(\beta/2)\sigma\tilde{F}_{|\ell|}(Q) + \boxed{e^{-i(\Delta J\phi_{\boldsymbol{Q}}+\alpha)}}\sin(\beta/2)\sigma'\tilde{F}_{|\ell'|}(Q)\right|^2$$




$$\left|\widetilde{M}_{G,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'}\right|^2 \approx \left(\frac{E_g D_{G}^{X}}{2\sqrt{2}\hbar}\right)^2\sin^2\theta_{\boldsymbol{Q}}\left|\cos(\beta/2)\tilde{F}_{|\ell|}(Q) + \boxed{e^{-i(\Delta J\phi_{\boldsymbol{Q}}+\alpha)}}\sin(\beta/2)\tilde{F}_{|\ell'|}(Q)\right|^2$$

$\boldsymbol{\mathcal{A}}^{\sigma,\ell,p;C}(\boldsymbol{q}_\parallel)$




$\boldsymbol{\mathcal{A}}^{\sigma',\ell',p;C}(\boldsymbol{q}_\parallel)$

$\Delta J = 0$ causing rotational symmetry in Q space

# Excitations of Excitons with Vector Vortex Beams

**Transition rate** of absorption $\Gamma$ root from the Fermi's golden rule, describe the optical transition as from many body ground state to the exciton state with center-of-mass momentum $\boldsymbol{Q}$

$$\Gamma_{S,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'} = \frac{2\pi}{\hbar}\left|\widetilde{M}_{S,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'}\right|^2 \rho\left(\hbar\omega = E_{S,\boldsymbol{Q}}^{X}\right)$$

$$M_{S,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'} \approx \cos(\beta/2)\, M_{S,\boldsymbol{Q}}^{\sigma,\ell} + e^{i\alpha}\sin(\beta/2)\, M_{S,\boldsymbol{Q}}^{\sigma',\ell'}$$

$$M_{S,\boldsymbol{Q}}^{\sigma,\ell} \approx \frac{E_g}{2i\hbar}\boldsymbol{\mathcal{A}}^{\sigma,\ell}(\boldsymbol{Q})\cdot \boldsymbol{D}_{S,\boldsymbol{Q}}^{X*}$$

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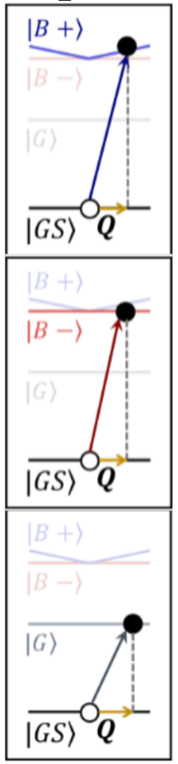
$$\boldsymbol{\mathcal{A}}^{\sigma,\ell,p;C}(\boldsymbol{q}_\parallel) \approx \tilde{F}_{|\ell|,p}(q_\parallel)e^{i\ell\phi_{\boldsymbol{q}}}\hat{\boldsymbol{\epsilon}} - \frac{\sin\theta_{\boldsymbol{q}}}{\sqrt{2}}\tilde{F}_{|\ell|,p}(q_\parallel)e^{i(\ell+\sigma)\phi_{\boldsymbol{q}}}\hat{\boldsymbol{z}}$$

$$\tilde{F}_{|\ell|,p=0}(q_\parallel) = \tilde{F}_{|\ell|}(Q) = (-i)^{|\ell|}\frac{A_0}{\Omega}\sqrt{\frac{2\pi}{|\ell|!}}W_0^2\left(\frac{q_\parallel W_0}{\sqrt{2}}\right)^{|\ell|}e^{-\frac{q_\parallel^2 W_0^2}{4}}$$

$$\boldsymbol{\mathcal{A}}^{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel) = \cos(\pi/4)\,\boldsymbol{\mathcal{A}}^{\sigma,\ell,p;C}(\boldsymbol{q}_\parallel) + e^{i\alpha=0}\sin(\pi/4)\,\boldsymbol{\mathcal{A}}^{\sigma',\ell',p;C}(\boldsymbol{q}_\parallel)$$

$$\Delta J = (\sigma' + \ell') - (\sigma + \ell)$$

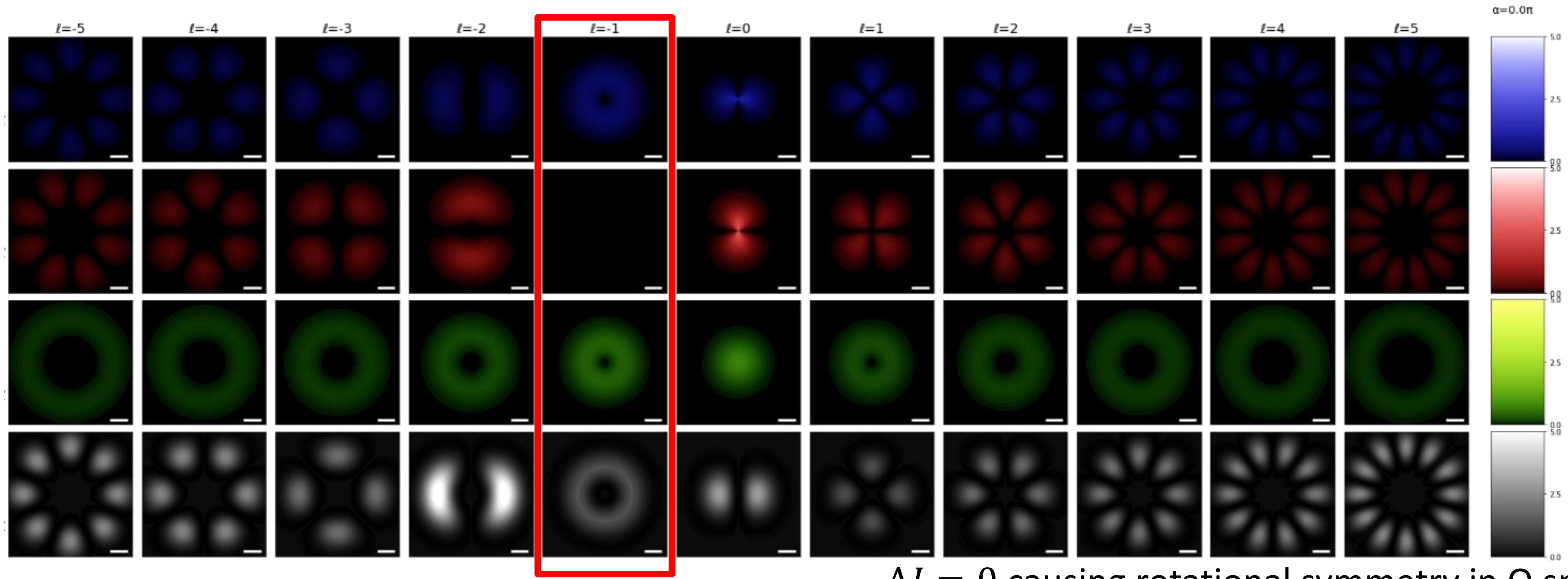
**Optical matrix elements for $p = 0$:**




$$\left|\widetilde{M}_{B+,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'}\right|^2 \approx$$

$$\left|\widetilde{M}_{B-,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'}\right|^2 \approx$$

$$\left|\widetilde{M}_{G,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'}\right|^2 \approx$$




$\Delta J = 0$ causing rotational symmetry in Q space

# Excitations of Excitons with Vector Vortex Beams

**Transition rate** of absorption Γ root from the Fermi's golden rule, describe the optical transition as from many body ground state to the exciton state with center-of-mass momentum $\boldsymbol{Q}$

$$\Gamma_{S,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'} = \frac{2\pi}{\hbar}\left|\widetilde{M}_{S,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'}\right|^2 \rho\left(\hbar\omega = E_{S,\boldsymbol{Q}}^{X}\right)$$

$$M_{S,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'} \approx \cos(\beta/2)\, M_{S,\boldsymbol{Q}}^{\sigma,\ell} + e^{i\alpha}\sin(\beta/2)\, M_{S,\boldsymbol{Q}}^{\sigma',\ell'}$$

$$M_{S,\boldsymbol{Q}}^{\sigma,\ell} \approx \frac{E_g}{2i\hbar}\boldsymbol{\mathcal{A}}^{\sigma,\ell}(\boldsymbol{Q})\cdot \boldsymbol{D}_{S,\boldsymbol{Q}}^{X*}$$

**Vector Vortex Beams** is composed by two twisted light, and we denote it with each light's spin $\sigma$ and topological charge $\ell$

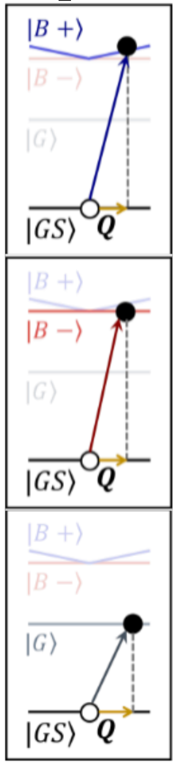
$$\boldsymbol{\mathcal{A}}^{\sigma,\ell,p;C}(\boldsymbol{q}_\parallel) \approx \tilde{F}_{|\ell|,p}(q_\parallel)e^{i\ell\phi_{\boldsymbol{q}}}\hat{\boldsymbol{\epsilon}} - \frac{\sin\theta_{\boldsymbol{q}}}{\sqrt{2}}\tilde{F}_{|\ell|,p}(q_\parallel)e^{i(\ell+\sigma)\phi_{\boldsymbol{q}}}\hat{\boldsymbol{z}}$$

$$\tilde{F}_{|\ell|,p=0}(q_\parallel) = \tilde{F}_{|\ell|}(Q) = (-i)^{|\ell|}\frac{A_0}{\Omega}\sqrt{\frac{2\pi}{|\ell|!}}W_0^2\left(\frac{q_\parallel W_0}{\sqrt{2}}\right)^{|\ell|}e^{-\frac{q_\parallel^2 W_0^2}{4}}$$

$$\boldsymbol{\mathcal{A}}^{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel) = \cos(\pi/4)\,\boldsymbol{\mathcal{A}}^{\sigma,\ell,p;C}(\boldsymbol{q}_\parallel) + e^{i\alpha=\frac{\pi}{2}}\sin(\pi/4)\,\boldsymbol{\mathcal{A}}^{\sigma',\ell',p;C}(\boldsymbol{q}_\parallel)$$

$$\Delta J = (\sigma' + \ell') - (\sigma + \ell)$$

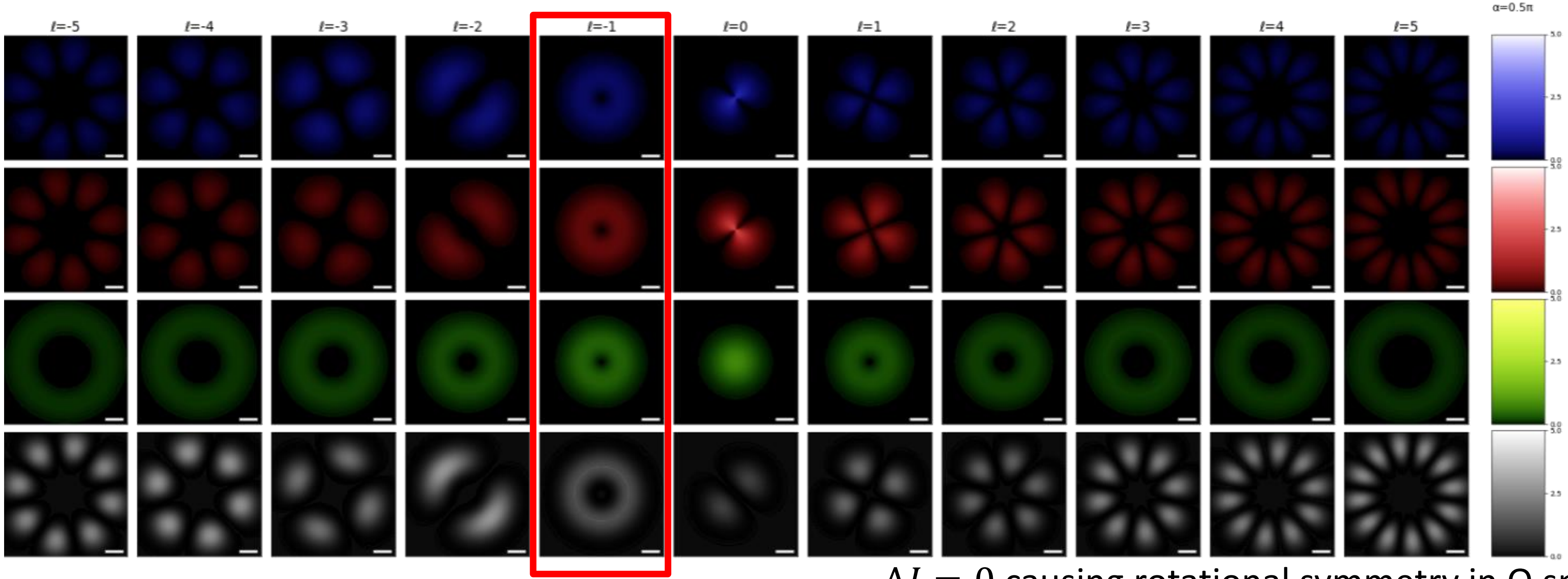
**Optical matrix elements for $p = 0$:**




$$\left|\widetilde{M}_{B+,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'}\right|^2 \approx$$

$$\left|\widetilde{M}_{B-,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'}\right|^2 \approx$$

$$\left|\widetilde{M}_{G,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'}\right|^2 \approx$$




$\Delta J = 0$ causing rotational symmetry in Q space

# Excitations of Excitons with Vector Vortex Beams

**Transition rate** of absorption Γ root from the Fermi's golden rule, describe the optical transition as from many body ground state to the exciton state with center-of-mass momentum $\boldsymbol{Q}$

$$\Gamma_{S,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'} = \frac{2\pi}{\hbar}\left|\widetilde{M}_{S,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'}\right|^2 \rho\left(\hbar\omega = E_{S,\boldsymbol{Q}}^{X}\right)$$

$$M_{S,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'} \approx \cos(\beta/2)\, M_{S,\boldsymbol{Q}}^{\sigma,\ell} + e^{i\alpha}\sin(\beta/2)\, M_{S,\boldsymbol{Q}}^{\sigma',\ell'}$$

$$M_{S,\boldsymbol{Q}}^{\sigma,\ell} \approx \frac{E_g}{2i\hbar}\boldsymbol{\mathcal{A}}^{\sigma,\ell}(\boldsymbol{Q})\cdot \boldsymbol{D}_{S,\boldsymbol{Q}}^{X*}$$

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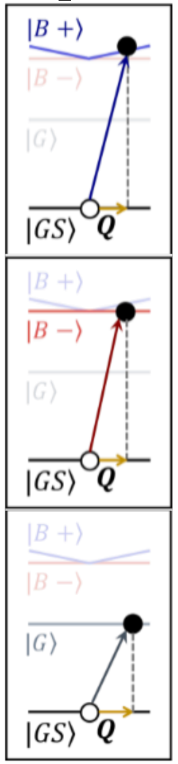
$$\boldsymbol{\mathcal{A}}^{\sigma,\ell,p;C}(\boldsymbol{q}_\parallel) \approx \tilde{F}_{|\ell|,p}(q_\parallel)e^{i\ell\phi_{\boldsymbol{q}}}\hat{\boldsymbol{\epsilon}} - \frac{\sin\theta_{\boldsymbol{q}}}{\sqrt{2}}\tilde{F}_{|\ell|,p}(q_\parallel)e^{i(\ell+\sigma)\phi_{\boldsymbol{q}}}\hat{\boldsymbol{z}}$$

$$\tilde{F}_{|\ell|,p=0}(q_\parallel) = \tilde{F}_{|\ell|}(Q) = (-i)^{|\ell|}\frac{A_0}{\Omega}\sqrt{\frac{2\pi}{|\ell|!}}W_0^2\left(\frac{q_\parallel W_0}{\sqrt{2}}\right)^{|\ell|}e^{-\frac{q_\parallel^2 W_0^2}{4}}$$

$$\boldsymbol{\mathcal{A}}^{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel) = \cos(\pi/4)\,\boldsymbol{\mathcal{A}}^{\sigma,\ell,p;C}(\boldsymbol{q}_\parallel) + e^{i\alpha=\pi}\sin(\pi/4)\,\boldsymbol{\mathcal{A}}^{\sigma',\ell',p;C}(\boldsymbol{q}_\parallel)$$

$$\Delta J = (\sigma'+\ell') - (\sigma+\ell)$$

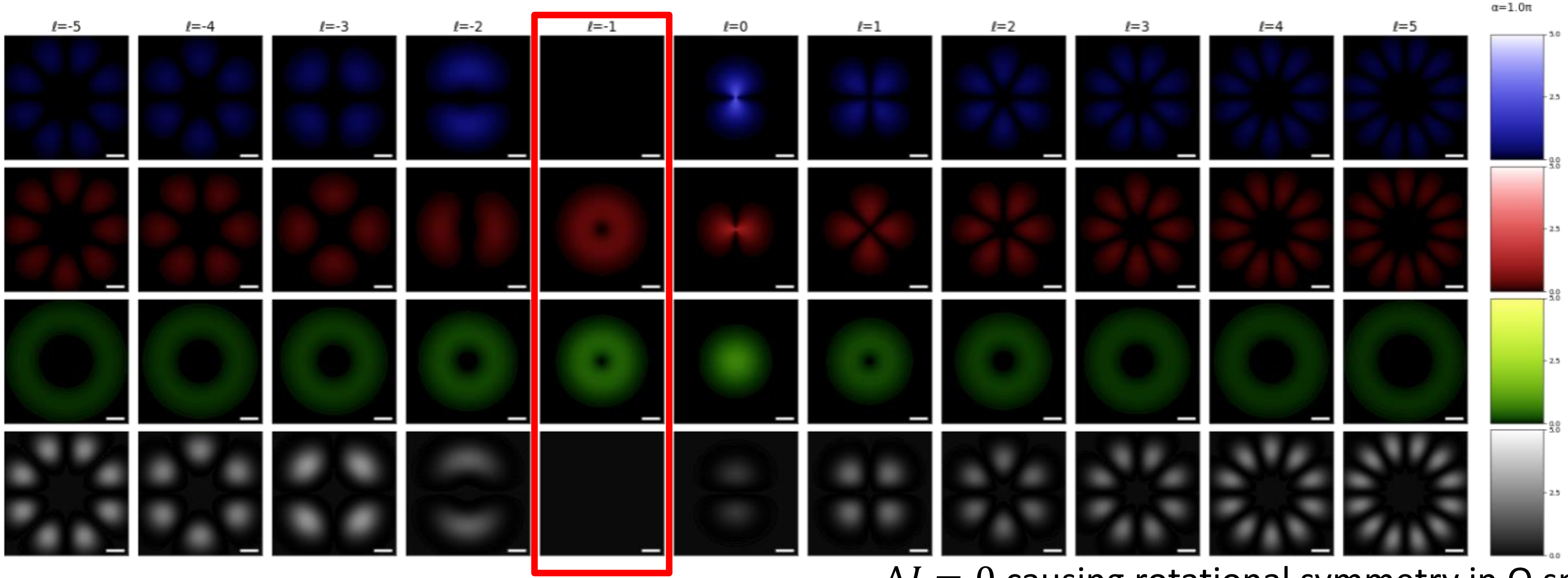
**Optical matrix elements for $p = 0$:**




$$\left|\widetilde{M}_{B+,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'}\right|^2 \approx$$

$$\left|\widetilde{M}_{B-,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'}\right|^2 \approx$$

$$\left|\widetilde{M}_{G,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'}\right|^2 \approx$$




$\Delta J = 0$ causing rotational symmetry in Q space

# Excitations of Excitons with Vector Vortex Beams

**Transition rate** of absorption $\Gamma$ root from the Fermi's golden rule, describe the optical transition as from many body ground state to the exciton state with center-of-mass momentum $\boldsymbol{Q}$

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$$M_{S,\boldsymbol{Q}}^{\sigma,\ell,\sigma',\ell'} \approx \cos(\beta/2)\, M_{S,\boldsymbol{Q}}^{\sigma,\ell} + e^{i\alpha}\sin(\beta/2)\, M_{S,\boldsymbol{Q}}^{\sigma',\ell'}$$

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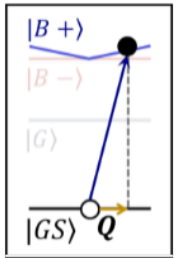
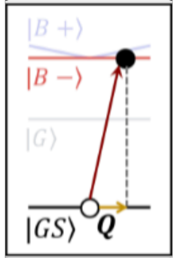
$$\boldsymbol{\mathcal{A}}^{\sigma,\ell,\sigma',\ell'}(\boldsymbol{q}_\parallel) = \cos(\beta/2)\,\boldsymbol{\mathcal{A}}^{\sigma,\ell,p;C}(\boldsymbol{q}_\parallel) + e^{i\alpha}\sin(\beta/2)\,\boldsymbol{\mathcal{A}}^{\sigma',\ell',p;C}(\boldsymbol{q}_\parallel)$$

$$\Delta J = (\sigma' + \ell') - (\sigma + \ell)$$



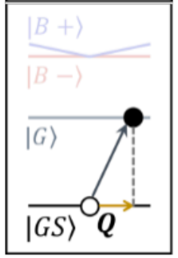

$$\Gamma_{B+}^{\sigma,\ell,\sigma',\ell'}(\alpha,\beta) \propto$$

$$\boxed{\frac{\Phi_e}{8\Omega}\left(\frac{E_g D_{B+}^X}{a_0\hbar}\right)^2\left\{1 + \delta_{\Delta J,0}\sin(\beta)\cos\left[\alpha - \frac{\pi}{2}(|\ell'| - |\ell|)\right]\frac{[(|\ell| + |\ell'|)/2]!}{\sqrt{|\ell| + |\ell'|!}}\right\}}$$




$$\Gamma_{B-}^{\sigma,\ell,\sigma',\ell'}(\alpha,\beta) \propto$$

$$\boxed{\frac{\Phi_e}{8\Omega}\left(\frac{E_g D_{B-}^X}{a_0\hbar}\right)^2\left\{1 + \delta_{\Delta J,0}\sin(\beta)\cos\left[\alpha - \frac{\pi}{2}(\sigma' + |\ell'| - \sigma - |\ell|)\right]\frac{[(|\ell| + |\ell'|)/2]!}{\sqrt{|\ell| + |\ell'|!}}\right\}}$$




$$\Gamma_{G}^{\sigma,\ell,\sigma',\ell'}(\alpha,\beta) \propto$$

$$\boxed{\frac{\Phi_e}{4\Omega}\left(\frac{E_g D_{G}^X}{a_0\hbar q_0 W_0}\right)^2\left\{|\ell|\cos^2\left(\frac{\beta}{2}\right) + |\ell'|\sin^2\left(\frac{\beta}{2}\right) + 1 + \delta_{\Delta J,0}\sin(\beta)\cos\left[\alpha - \frac{\pi}{2}(|\ell'| - |\ell|)\right]\frac{[(|\ell| + |\ell'|)/2 + 1]!}{\sqrt{|\ell| + |\ell'|!}}\right\}}$$

$\Gamma_{B\pm}^{1,\ell,-1,-\ell}(\alpha,\beta=\pi/2)$ $\Gamma_{G}^{1,\ell,-1,-\ell}(\alpha,\beta=\pi/2)$